Question Paper Code 25900419

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Value Addition Course (VAC)- MG3VACGEO200 - SUSTAINABLE RESOURCE MANAGEMENT

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

ANSWER KEY

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer in 60 words
Answer any **7** questions
Each question carries **2** marks

1. Name two perennial rivers in India. [K] / [CO1]

Ganga, Brahmaputra, Krishna, Godavari, Periyar (Any two-1 mark)

2. State advantages of tidal energy. [K] / [CO2]

Predictable, renewable, no greenhouse gas emission /long lifespan. Any two (1 mark each)

3. State the importance of integrated approach in sustainability. [An] / [CO3]

Balances environmental, economic, and social priorities, provides holistic solutions/long-term benefits (Any two importance -2 mark)

4. Define tidal energy. [K] / [CO4]

Tidal energy refers to energy from ocean tides-used for power generation/renewable energy (Definition- 2 mark)

5. Explain non-conventional resource. Give examples. [U] / [CO1]

Definition (1 mark); Examples: solar energy, wind energy (Any two -1 mark)

6. Explain soil degradation. [U] / [CO2]

Decline in soil quality (Definition-1 mark); Causes or effects (1 mark)

7. Explain global warming. [U] / [CO3]

Gradual increase in Earth's average temperature (Definition 1 mark); Explanation (1 mark)

8. Mention any four human impacts on natural resources. [K] / [CO1]

Deforestation, overgrazing, pollution, excessive mining. Any 4 impacts (0.5 each)

Part B

Short notes in 250 words
Answer any 3 questions
Each question carries 8 marks

9. Explain renewable energy sources as a method of energy conservation. [U] / [CO1]

Definition of renewable energy (2 marks); Examples: solar, wind, tidal, wave or water (Any two-2 marks); Environmental benefits and conservation impact (4 marks)

10. Explain renewable energy resources. Write a note on merits and demerits of renewable energy. [U] / [CO2]

Definition of renewable energy (2 marks); Examples: solar, wind, tidal, wave or water (Any two-2 marks); Merits- Sustainability/environmental benefit (2 marks)- Demerits-Variability/high initial cost (2 marks)

11. Give an account of type of forest resources. [U] / [CO2]

Definition of forest resources (2 marks); Main types: timber products (3 marks); Non-timber products/ecological services (3 marks)

12. Explain the concept of Integrated Resource Management. [An] / [CO3]

Definition and core concept (4 marks); Examples or importance of integrated resource management (4 marks)

13. Define wind energy programme and explain its uses. [A] / [CO4]

Definition (4 marks); Uses-electricity generation, water pumping, rural electrification (Any 2 uses- 4 marks)

Part C
Essay in 400 words
Answer any 1 questions
Each question carries 12 marks

14. Discuss the impacts of excessive use of non-renewable resources. [U] / [CO1]

Definition of non-renewable resources (2 marks); Resource depletion, pollution and environmental impact, habitat loss. (Any four impacts- 8 marks); Sustainable practices, mitigation measures or related facts. (2 marks)

15. Explain the methods of sustainable water management and their significance. [U] / [CO2]

Definition of sustainable water management (2 marks); Significance (2 marks); Methods of sustainable water management- Rainwater harvesting, artificial recharge, watershed management, efficient irrigation, pollution control, recycling greywater (any 4 methods - 8 marks)

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